Geometry-based Methods in Vision

CMU 16-822, Fall 2024

Instructor: Shubham Tulsiani

Administrivia

- Assignment 1
  - *Due today*

- Problem Set 2 & Assignment 2 out
  - *Due in 1 and 2 weeks respectively*

Lecture 9: Two-view Geometry

Image credits: Noah Snavely

A reminder to ask questions!

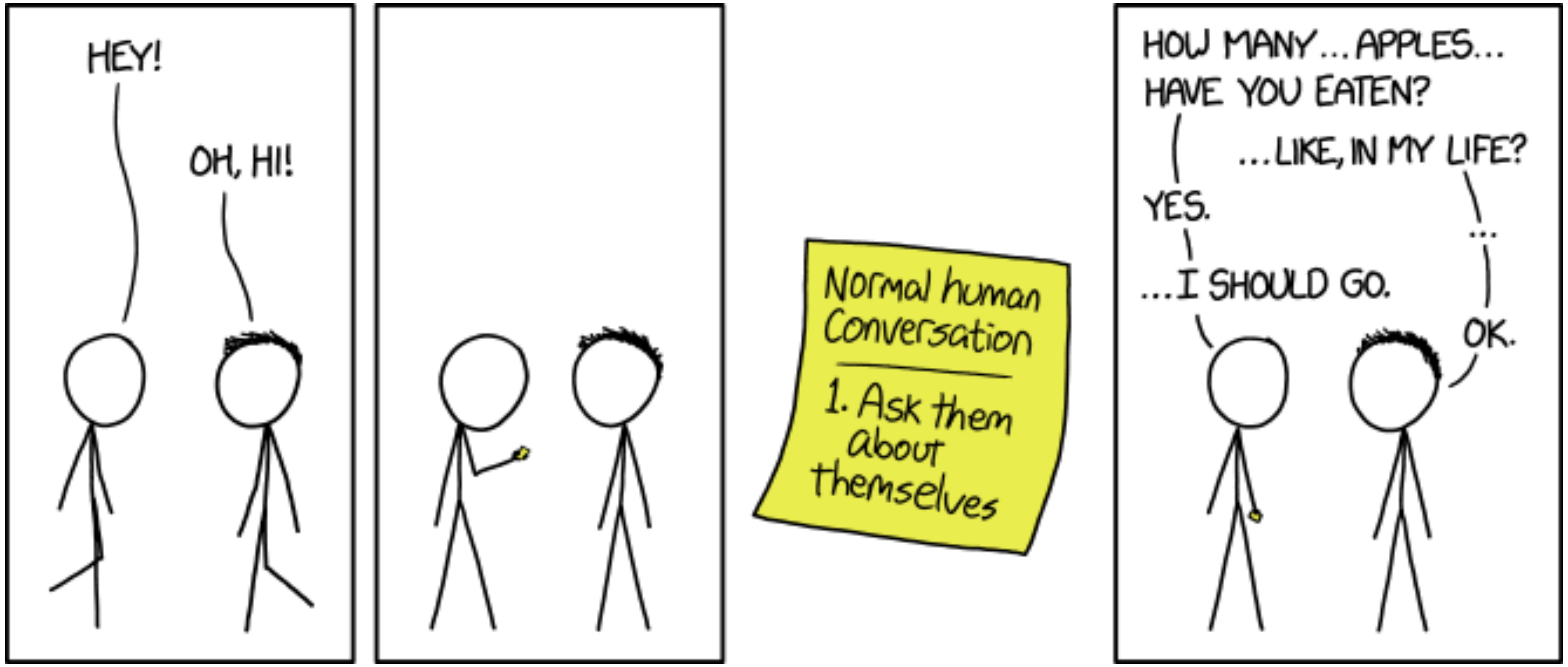


Image credits: xkcd

Single-view Reconstruction

Image credits: https://www.robots.ox.ac.uk/~vgg/projects/SingleView/

Single-view Reconstruction

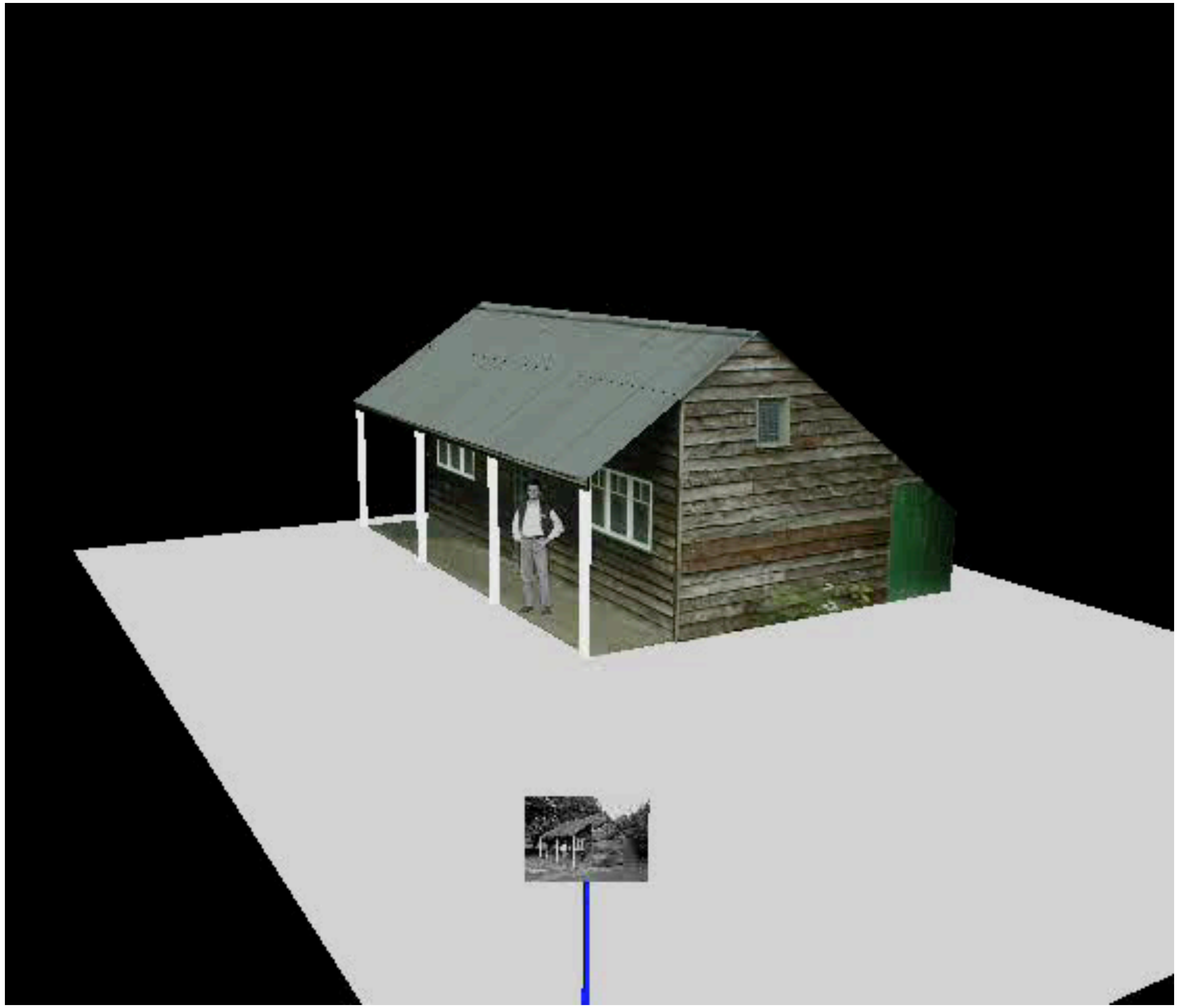
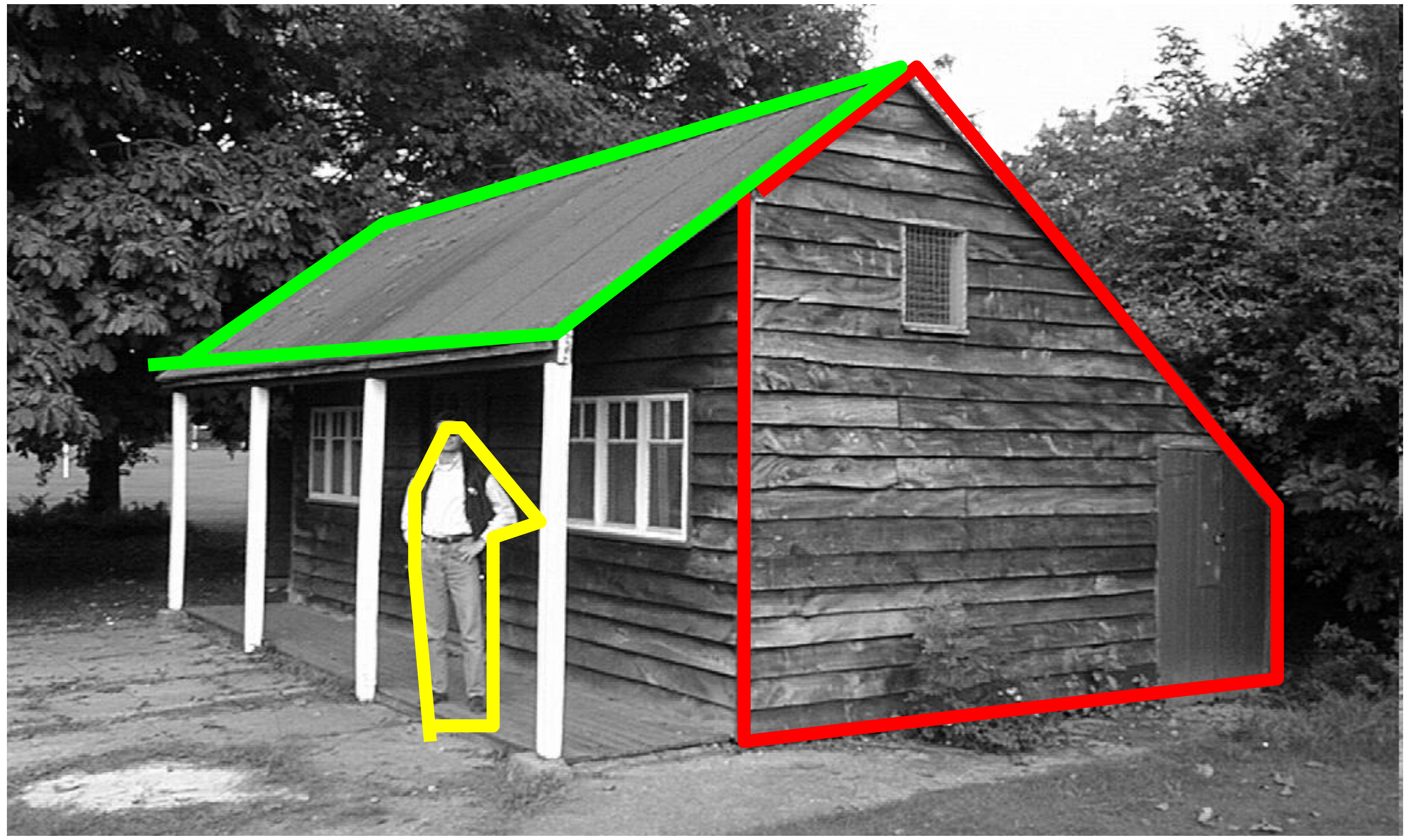
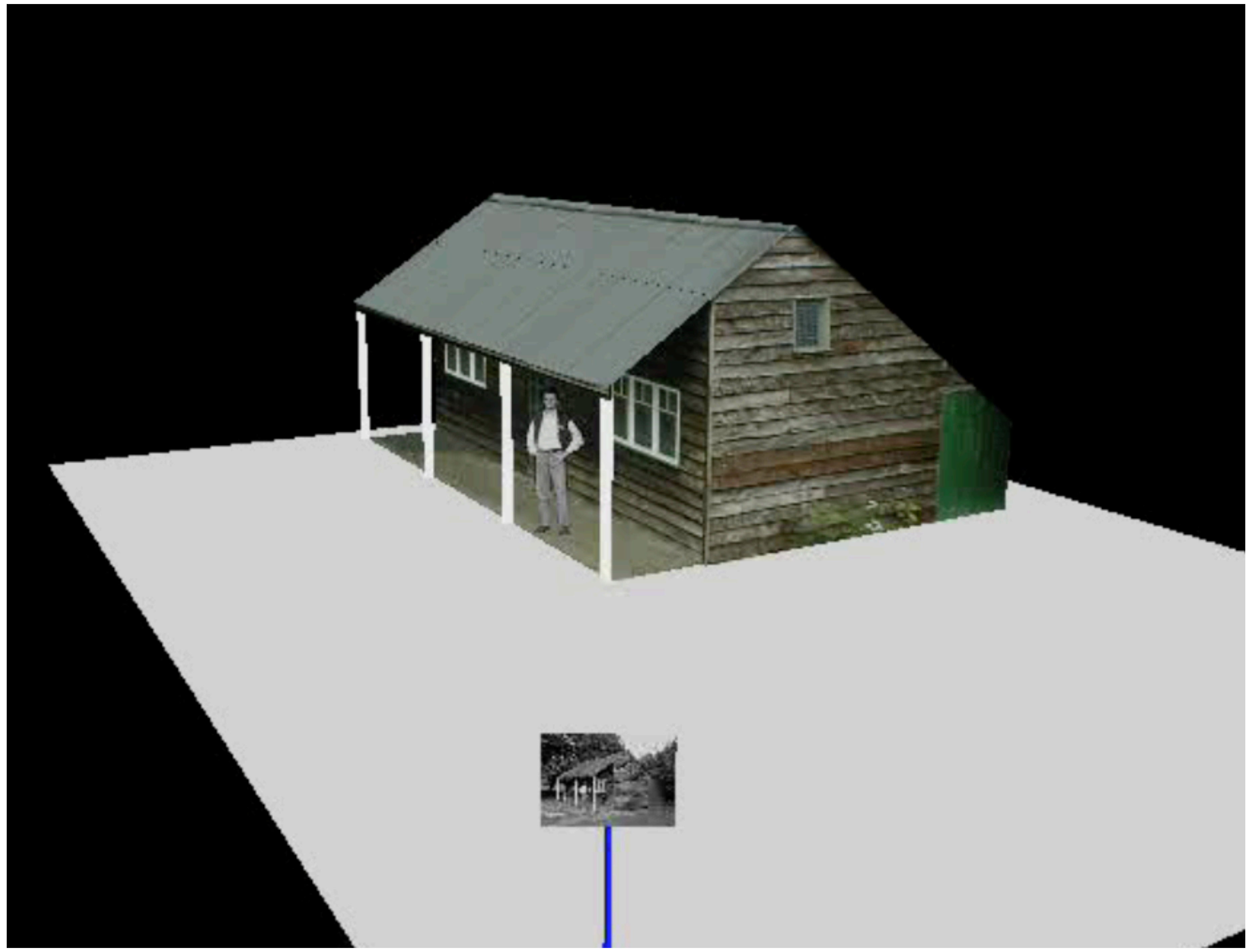


Image credits: https://www.robots.ox.ac.uk/~vgg/projects/SingleView/

Single-view Reconstruction





Possible to extract 3D from a single view using manual annotations

But, is it possible automatically?

and without priors about the world?

3D from Single-view — an Ill-posed Task

Fundamental ambiguities exist — cannot be geometrically resolved

Image credits: Lana Lazebnik

3D from Single-view — an Ill-posed Task

Video credits: Wikipedia

Multi-view to the rescue!

Multiple images of the same scene can resolve ambiguities

Disclaimer: not all of them!

Video credits: Wikipedia

Two-view Geometry

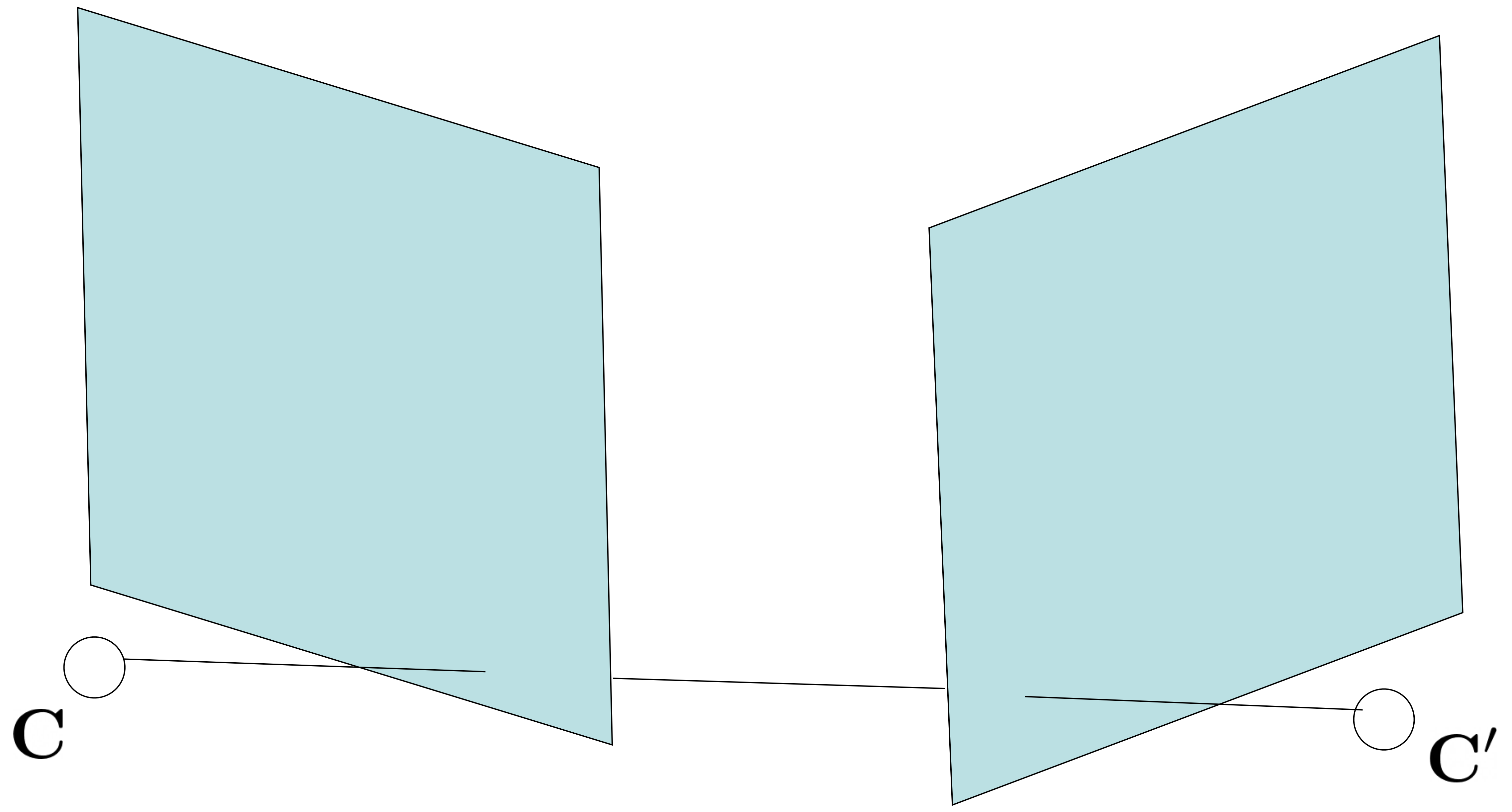
A stepping stone to 3D from multi-view

Develop concepts about cameras from correspondences, ambiguities in 3D and pose etc.

Image credits: Noah Snavely

Two-view Geometry: Setup and Notations

Two cameras observing a
common 3D world



$$\mathbf{P} \qquad \mathbf{P}'$$

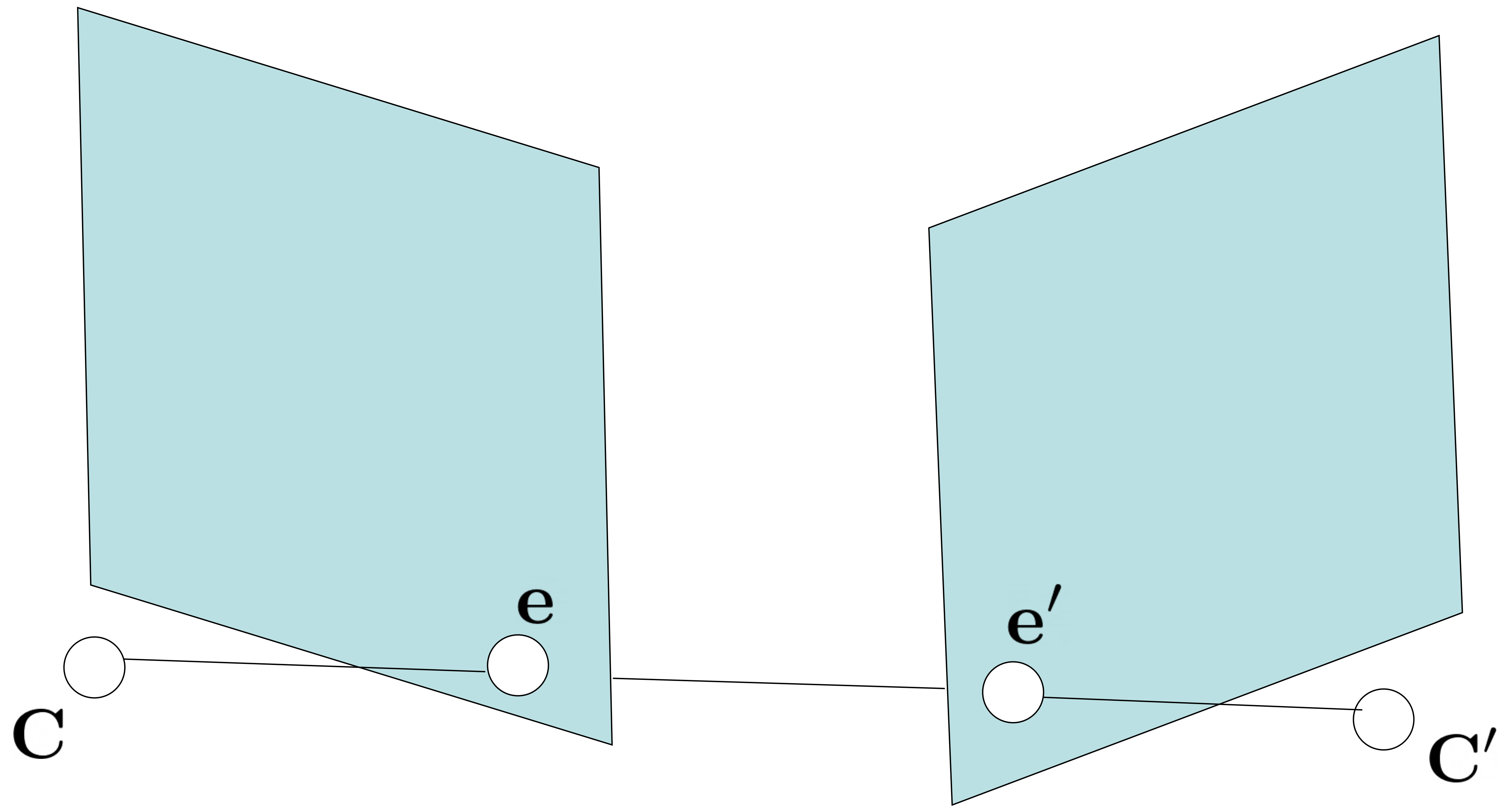
$$K, R, \tilde{\mathbf{C}} \qquad K', R', \tilde{\mathbf{C}}'$$

with corresponding projection
matrices

Image credits: Martial Hebert

Two-view Geometry: Setup and Notations

Two cameras observing a
common 3D world



$$\mathbf{P} \qquad \mathbf{P}'$$

$$K, R, \tilde{\mathbf{C}} \qquad K', R', \tilde{\mathbf{C}}'$$

with corresponding projection
matrices

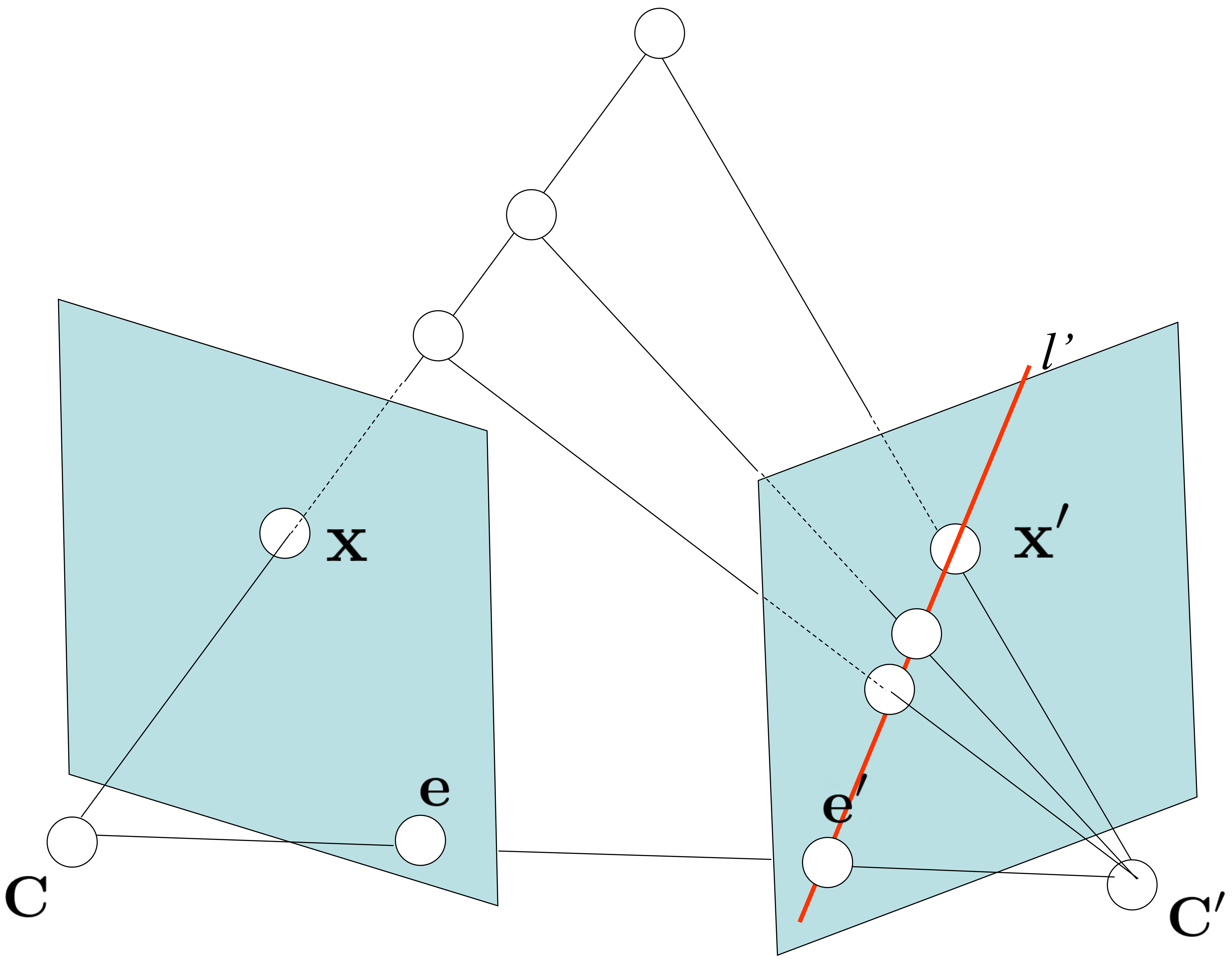
Epipoles

$$\mathbf{e} = \mathbf{P}\mathbf{C}' \qquad \mathbf{e}' = \mathbf{P}'\mathbf{C}$$

Projection of one camera centre
onto the other's image plane

Image credits: Martial Hebert

Correspondence across views



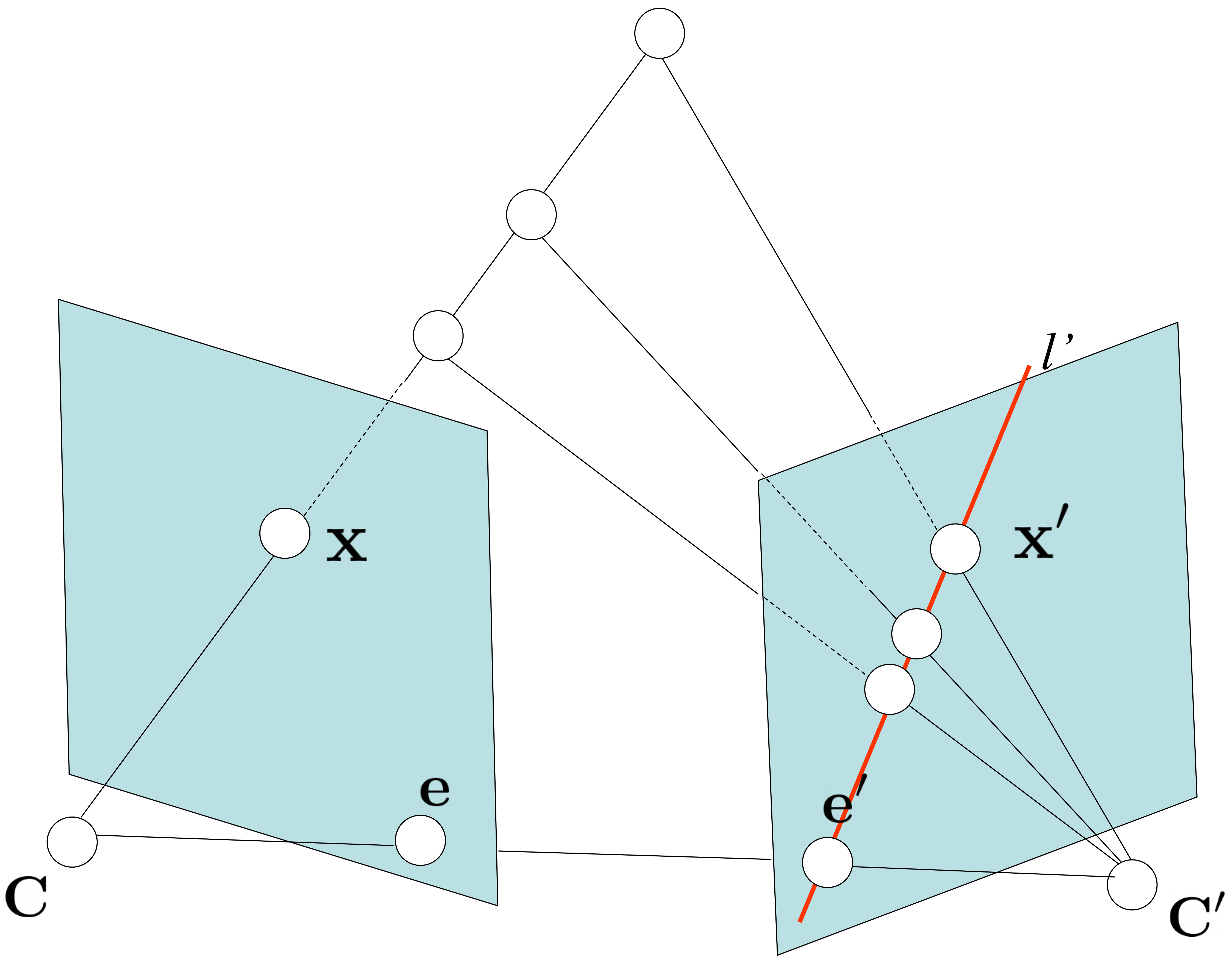
Observation 1: Given **x**, **x’** lies on a line

Observation 2: This line **always** passes the epipole **e’**

Q: Given an observed point in one view, where can the corresponding point be in the other?

Image credits: Martial Hebert

Fundamental Matrix — Mapping points to lines



$$\mathbf{l}' = F\mathbf{x}$$

Claim: A matrix allows computing the corresponding line **l'** via linear operation on **x**

Called the **Fundamental matrix**

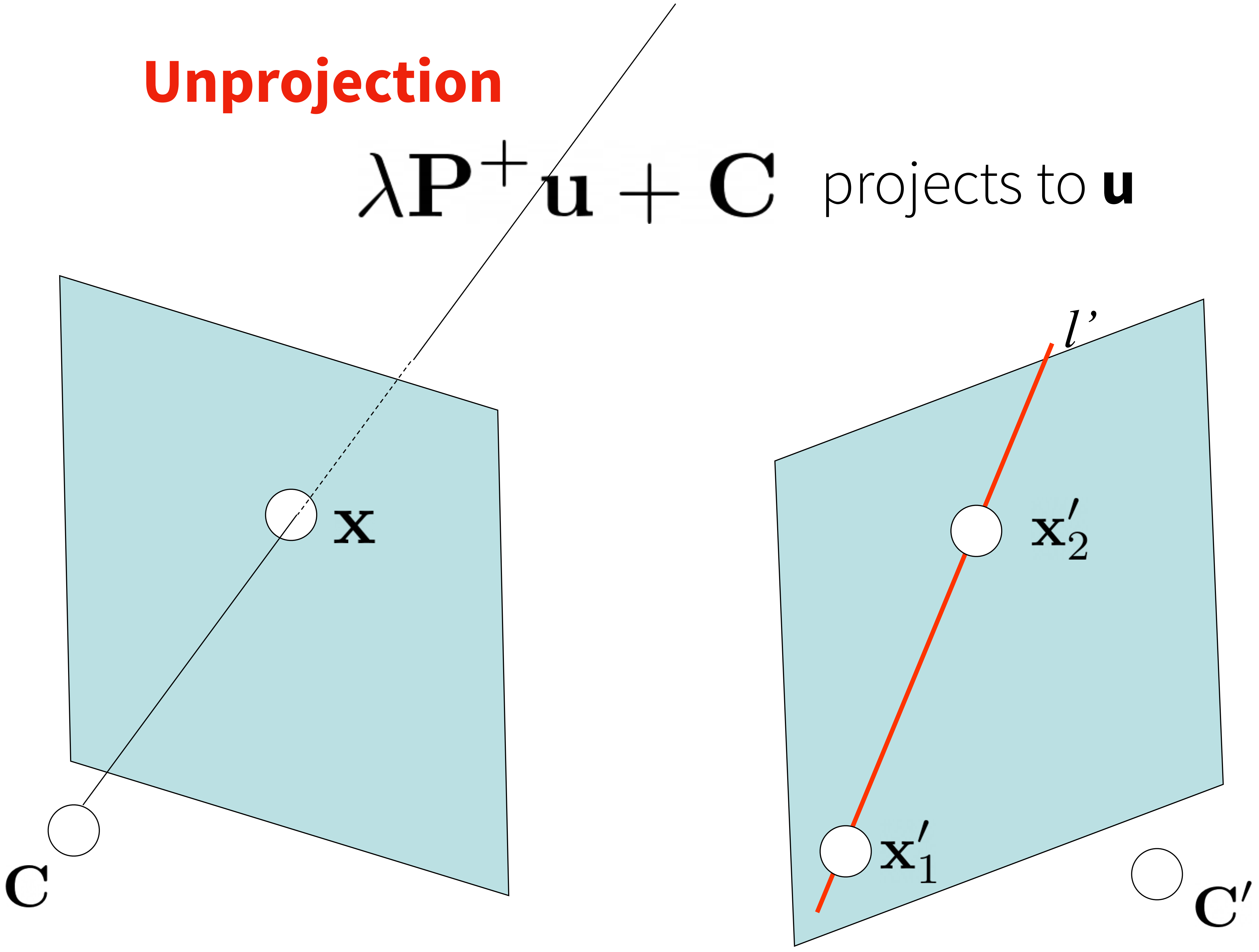
But, why is this a linear operation on **x**?

Image credits: Martial Hebert

Fundamental Matrix

Unprojection

$\lambda \mathbf{P}^{+}\mathbf{u} + \mathbf{C}$ projects to **u**



Given two possible projections:

$$\mathbf{l}' = \mathbf{x}'_1 \times \mathbf{x}'_2$$

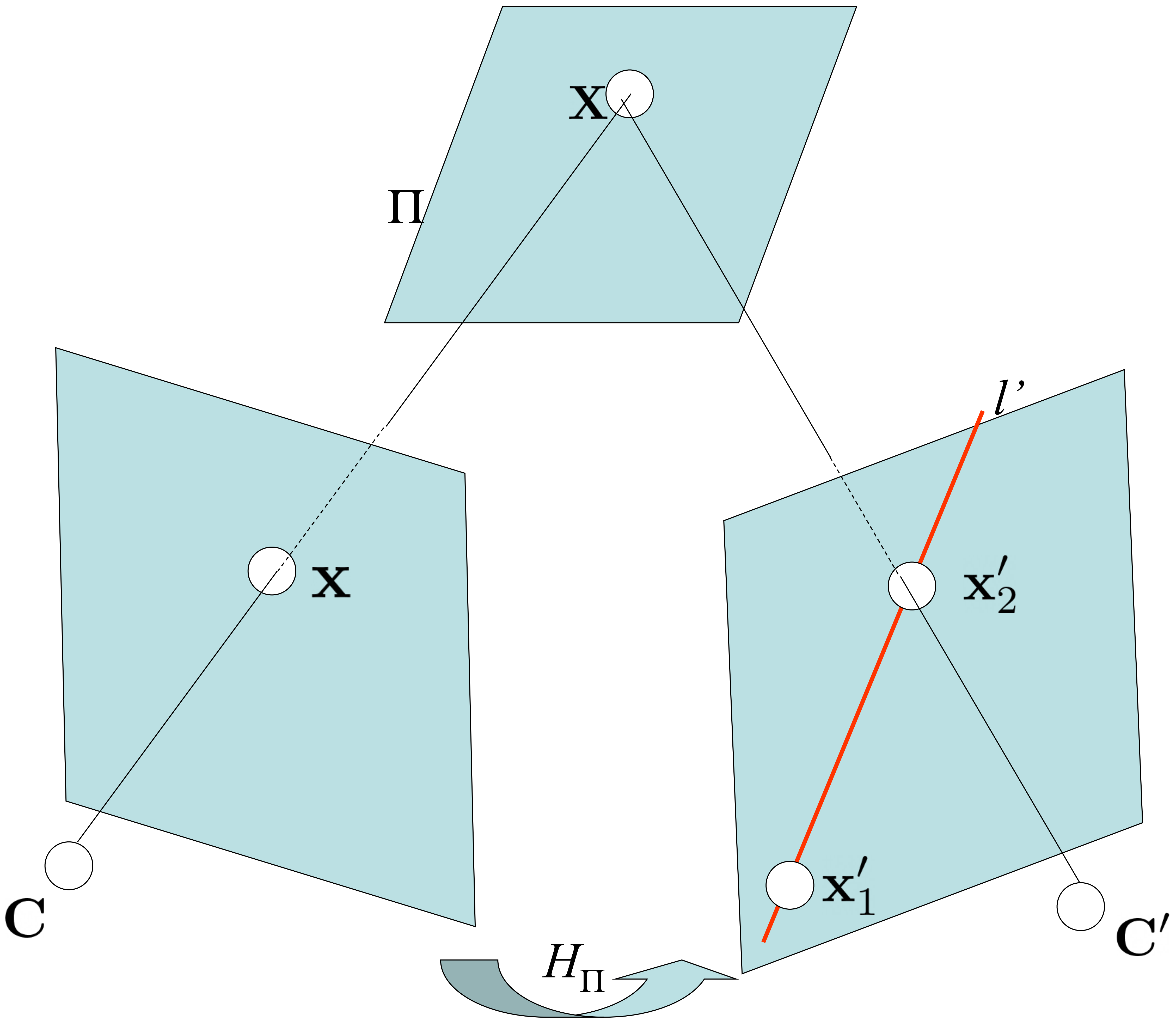
$$\mathbf{x}'_1 = \mathbf{P}'\mathbf{C} = \mathbf{e}' \qquad \mathbf{x}'_2 = \mathbf{P}'\mathbf{P}^{+}\mathbf{x}$$

$$\begin{aligned} \mathbf{l}' &= \mathbf{e}' \times \mathbf{P}'\mathbf{P}^{+}\mathbf{x} \\ &= \underbrace{[\mathbf{e}']_{\times}\mathbf{P}'\mathbf{P}^{+}}_{\text{Fundamental Matrix}}\mathbf{x} \end{aligned}$$

Q: Why is the line of possible projections **l'** linearly dependent on **x**?

Image credits: Martial Hebert

Fundamental Matrix via Planar Homography



Given two possible projections:

$$\mathbf{l}' = \mathbf{x}'_1 \times \mathbf{x}'_2$$

$$\mathbf{x}'_1 = \mathbf{P}'\mathbf{C} = \mathbf{e}' \qquad \mathbf{x}'_2 = H_\pi \mathbf{x}$$

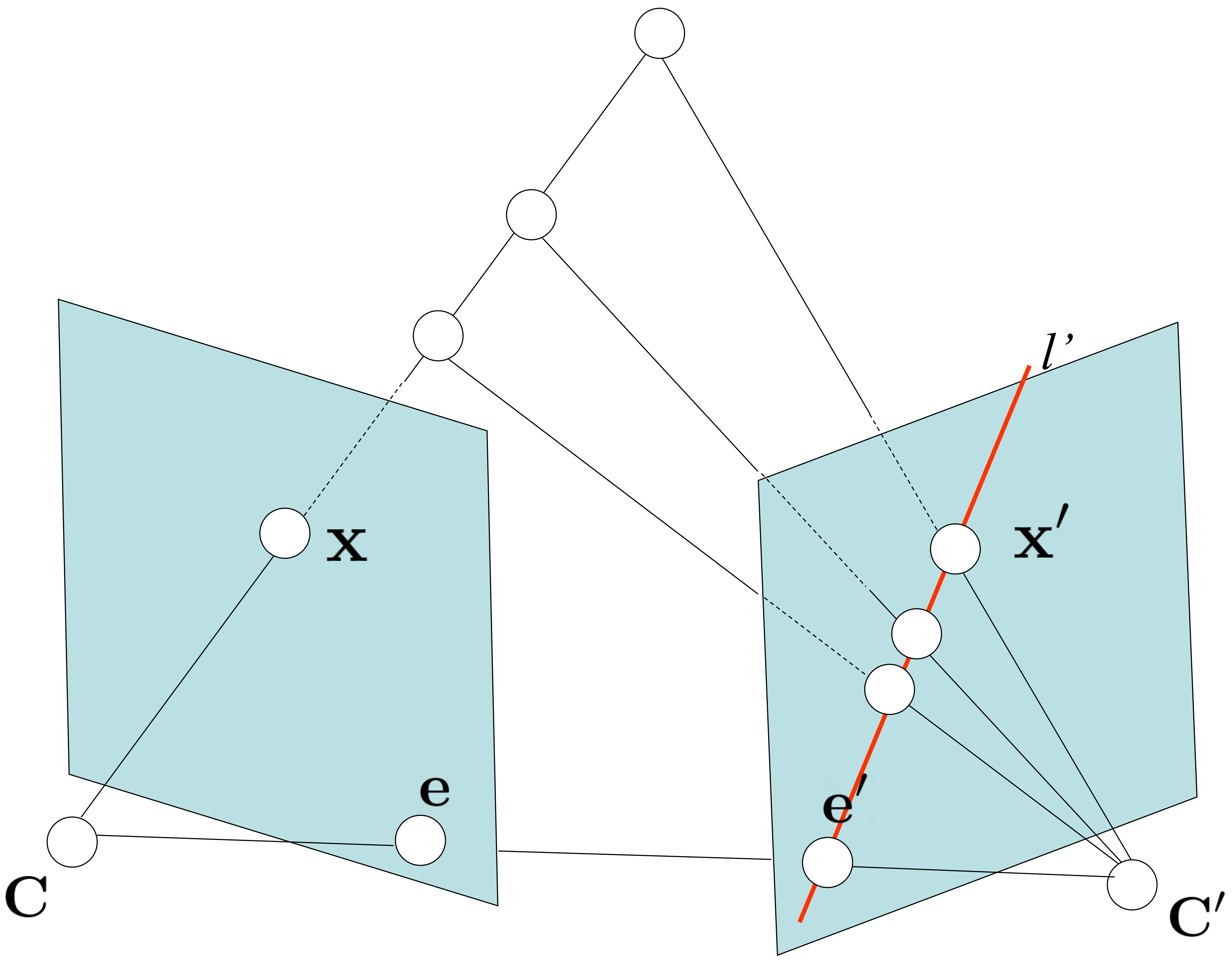
$$\mathbf{l}' = \underbrace{[\mathbf{e}']_\times H_\pi}_{\text{Fundamental Matrix}} \mathbf{x}$$

Note: not all 2D homographies correspond to planar homographies given the fundamental matrix (relevant later)

Let's consider the 3D point along the ray intersecting an arbitrary plane

Image credits: Martial Hebert

Fundamental Matrix — Mapping Points to Lines



Fundamental Matrix

$$\mathbf{l}' = F\mathbf{x}$$

We do not get a valid line for **x = e**

$$F\mathbf{e} = \mathbf{0}$$

Q: what is the rank of *F* ?

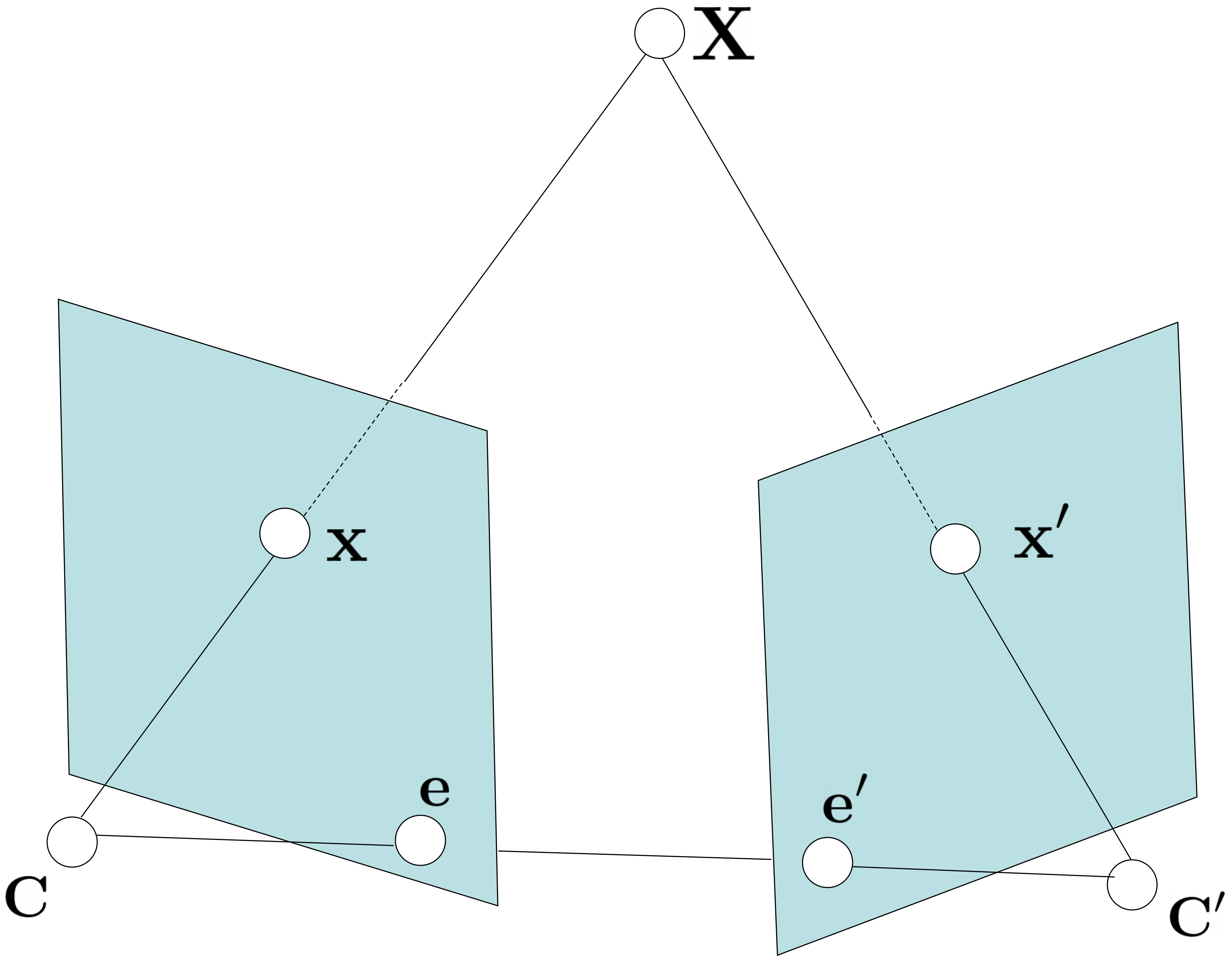
$$F = \begin{bmatrix} \mathbf{f}_1 & \mathbf{f}_2 & \alpha\mathbf{f}_1 + \beta\mathbf{f}_2 \end{bmatrix}$$

Q: Degrees of freedom ?

7 (defined up to a scale)

Image credits: Martial Hebert

Fundamental Matrix — Alternate view



Fundamental Matrix

$$\mathbf{l}' = F\mathbf{x}$$

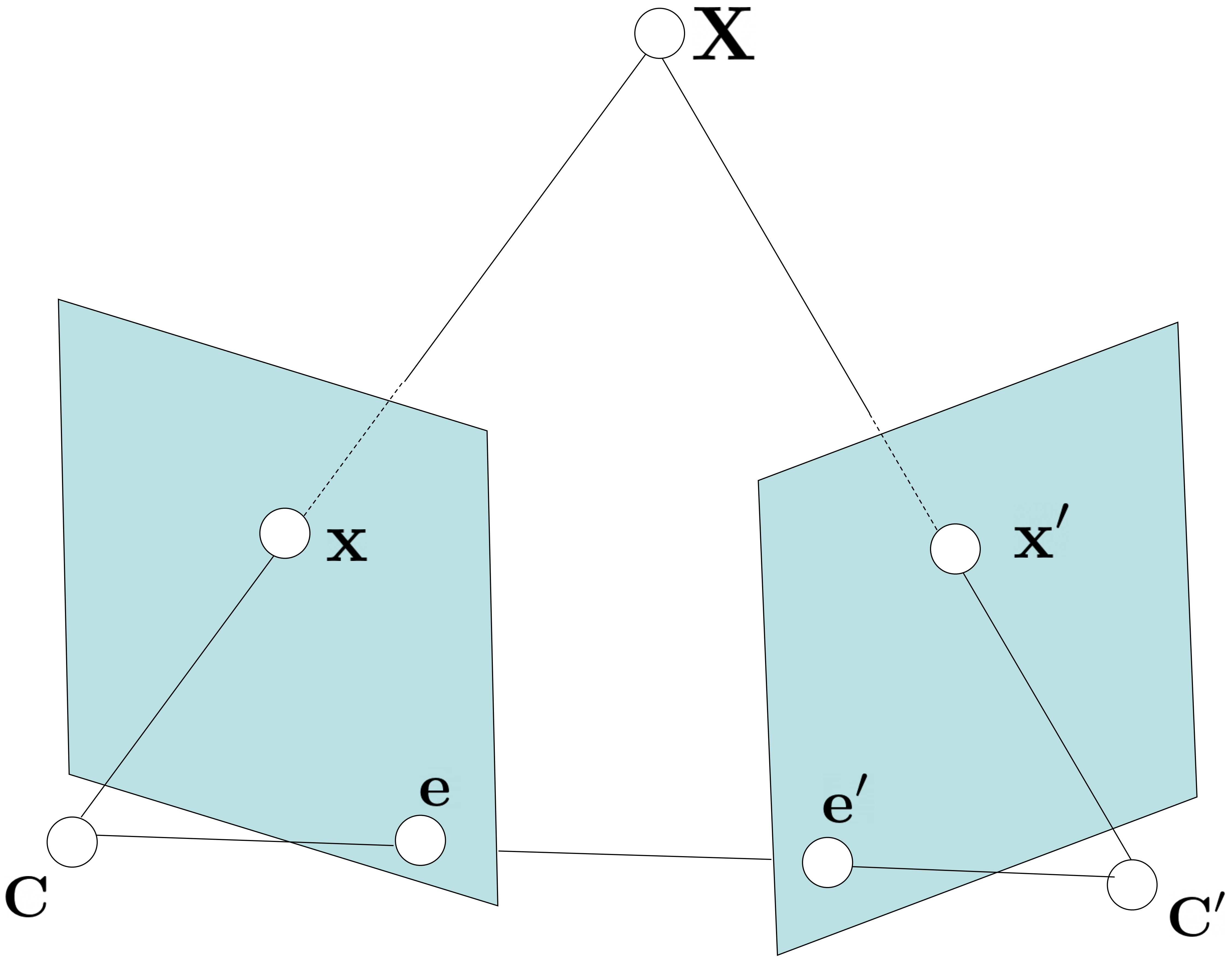
Alternate view:

$$\mathbf{x}'^{\mathsf{T}} F\mathbf{x} = 0$$

If (**x, x'**) are corresponding points (i.e. projection of same 3D point **X**), they satisfy the above

Image credits: Martial Hebert

Fundamental Matrix — Alternate view



$$\mathbf{x}'^{\top} F \mathbf{x} = 0$$

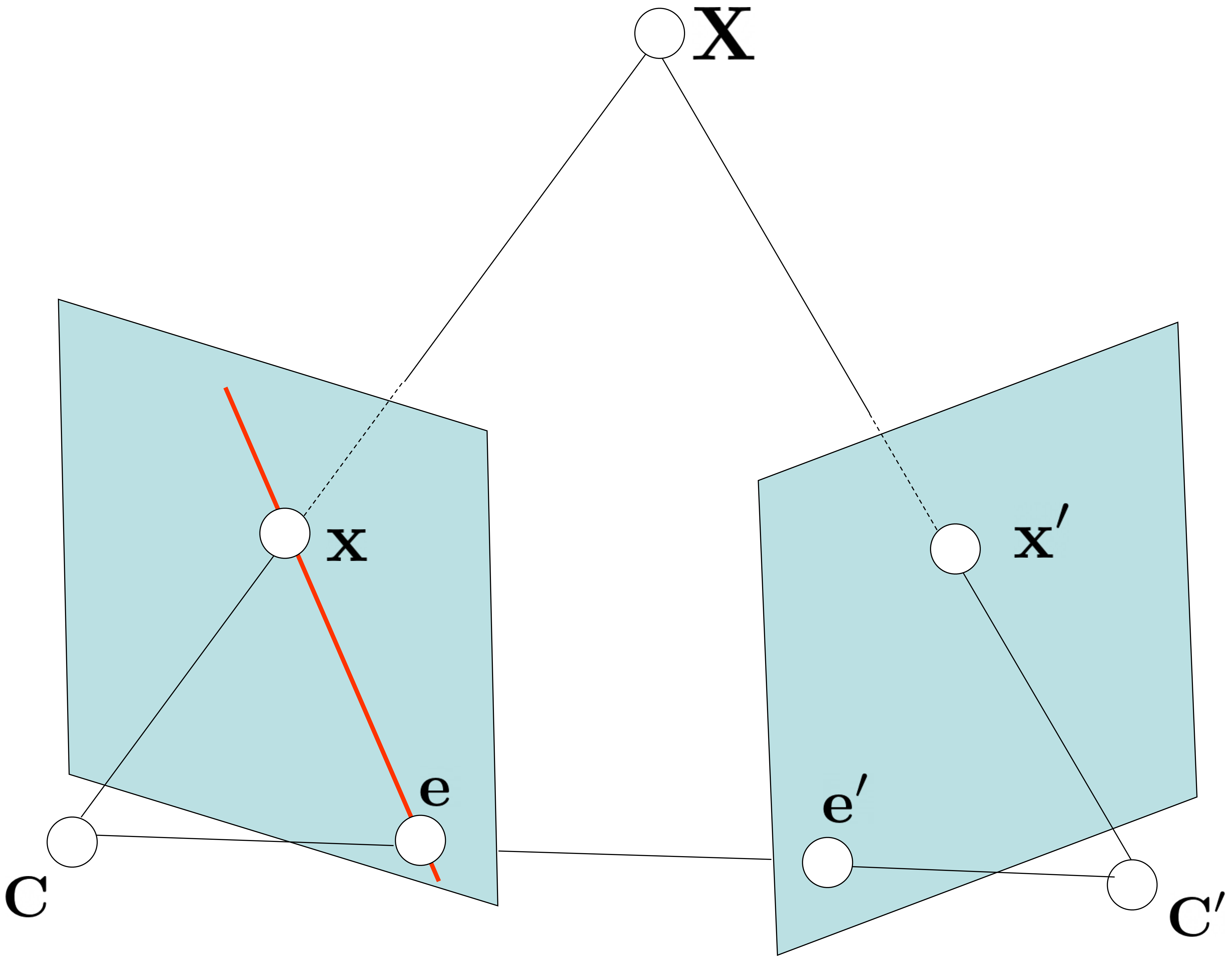
$$\forall \mathbf{X}, \quad \mathbf{X}^{\top} \underbrace{\mathbf{P}'^{\top} F \mathbf{P}}_{\text{Skew symmetric}} \mathbf{X} = 0$$

F is the Fundamental matrix between two cameras if and only if

$$\mathbf{P}'^{\top} F \mathbf{P}$$

is skew symmetric

Fundamental Matrix — Symmetry



$$\mathbf{x}'^{\mathsf{T}} F \mathbf{x} = 0 \qquad \mathbf{l}' = F\mathbf{x}$$

$$\text{if} \quad (\mathbf{P}, \mathbf{P}') \rightarrow F$$
$$\text{then} \quad (\mathbf{P}', \mathbf{P}) \rightarrow F^{\mathsf{T}}$$

$$\mathbf{l} = F^{\mathsf{T}} \mathbf{x}'$$

$$\mathbf{e}'^{\mathsf{T}} F = \mathbf{0}^{\mathsf{T}}$$

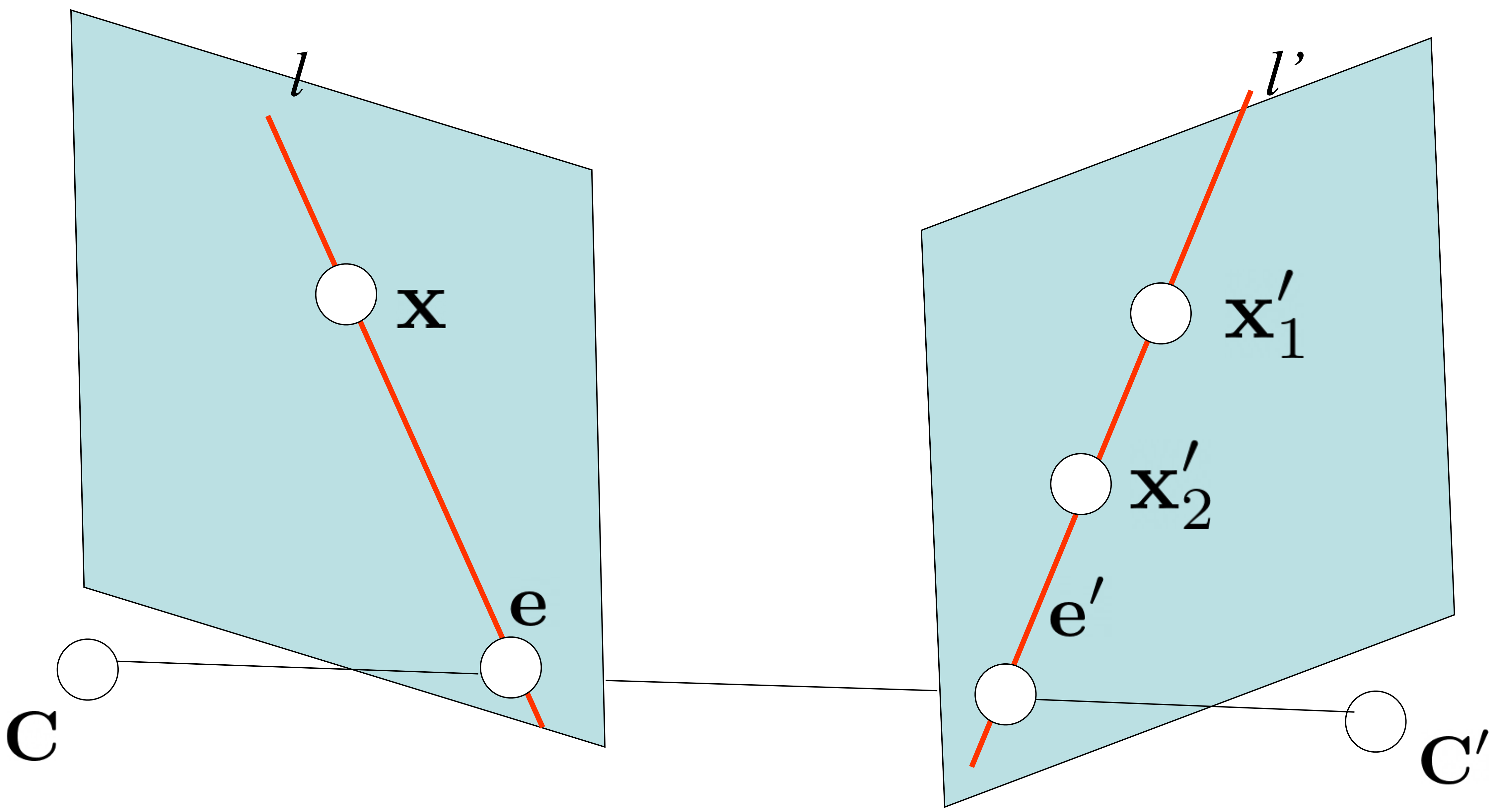
Visualizing Epipolar Lines

https://www.cs.cornell.edu/courses/cs5670/2022sp/demos/FundamentalMatrix/?demo=demo1

For all **x** along **l**, the epipolar line **l**’ is the same!

Q: Why do epipolar lines correspond?

Corresponding Epipolar Lines



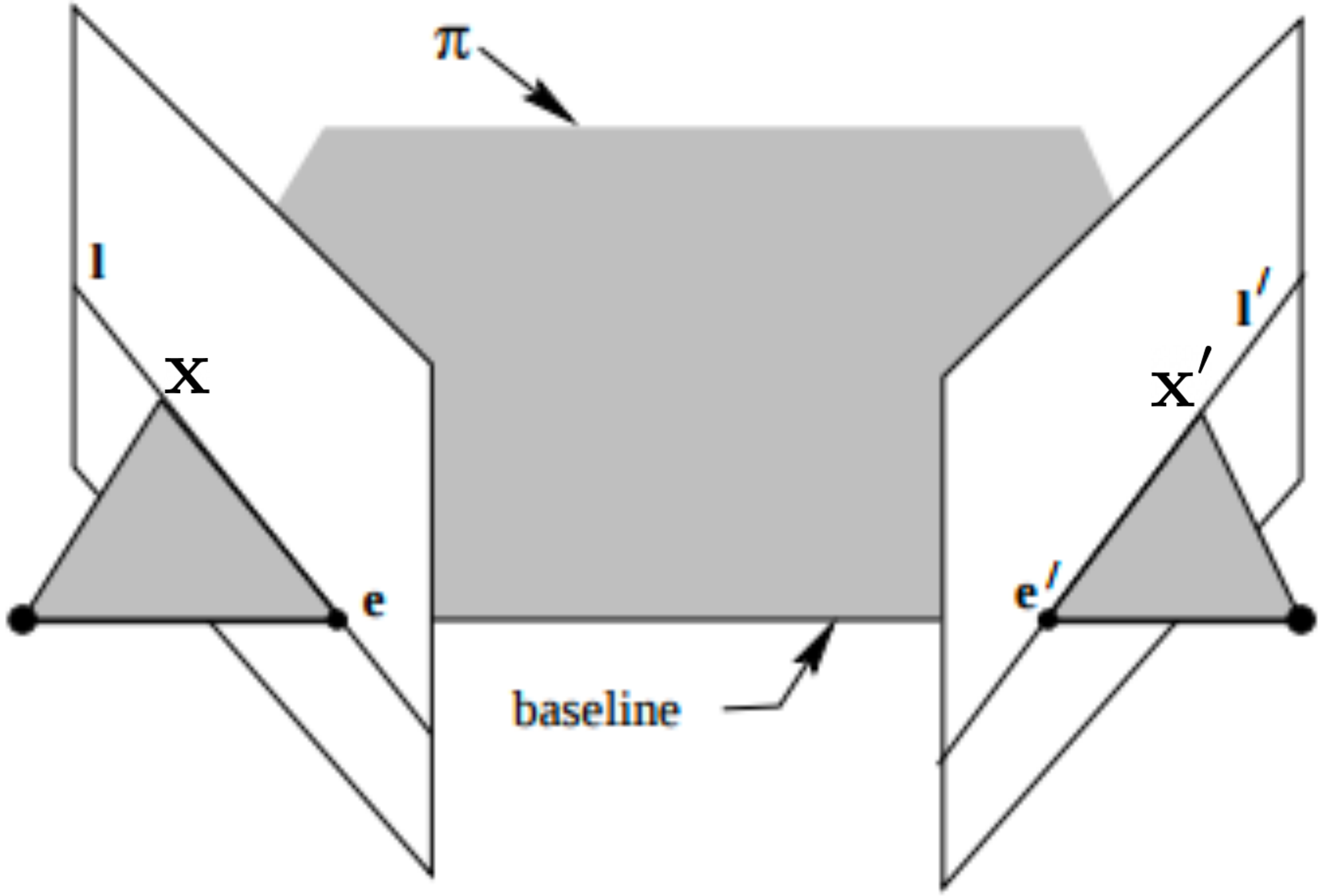
Say, $\mathbf{x}'_1$ and $\mathbf{x}'_2$ lie on $l' = F\mathbf{x}$

$$\mathbf{l}_1 = F^\top \mathbf{x}'_1 \qquad \mathbf{l}_2 = F^\top \mathbf{x}'_2$$

To prove : $\mathbf{l}_1 = \mathbf{l}_2 = \mathbf{l}$

Hint: find 2 points that lie on both $\mathbf{l}_1$, $\mathbf{l}_2$

Epipolar Planes and Corresponding Lines



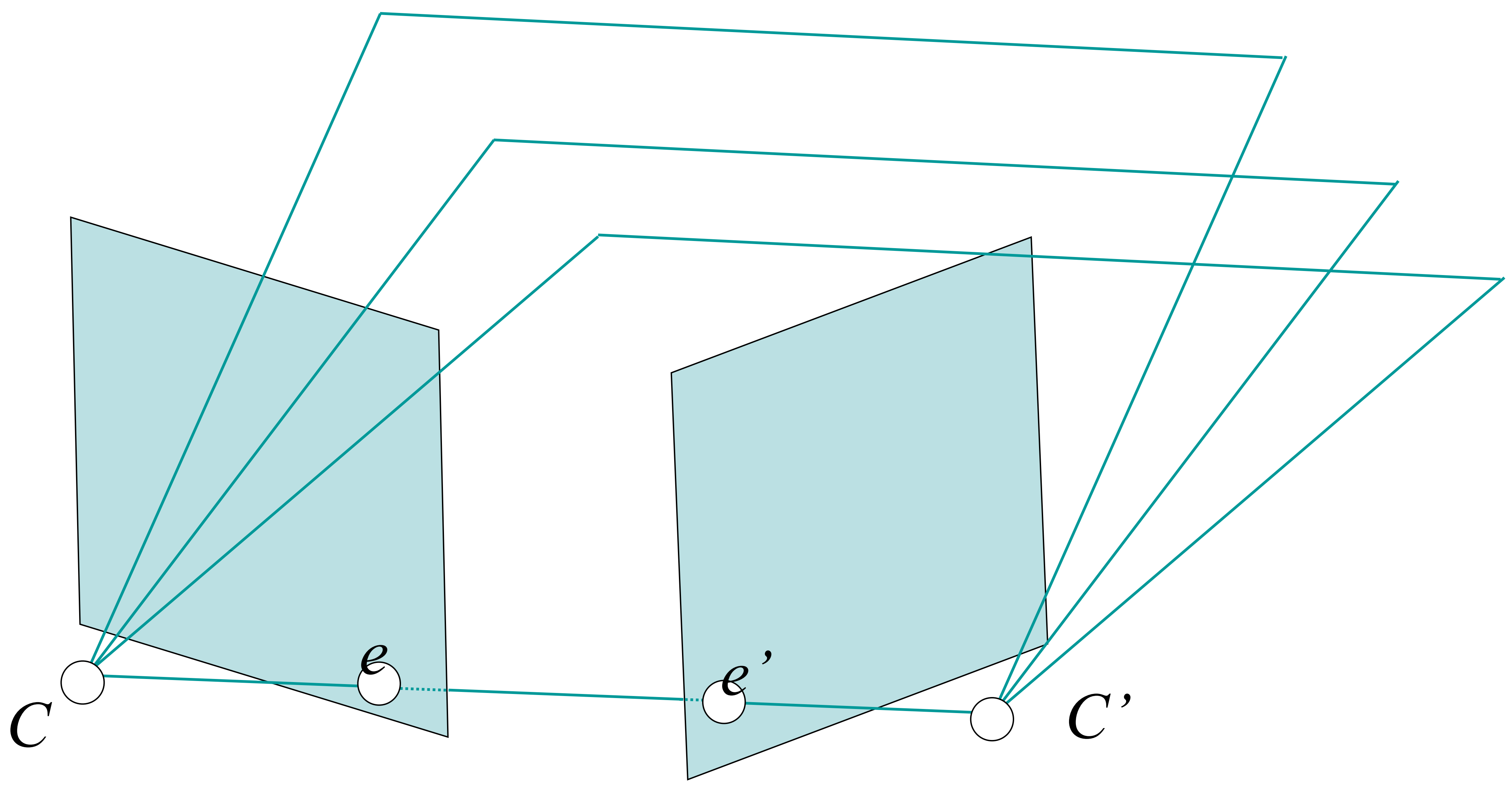
Consider the plane passing **C**, **C**', (**x** or **x**')

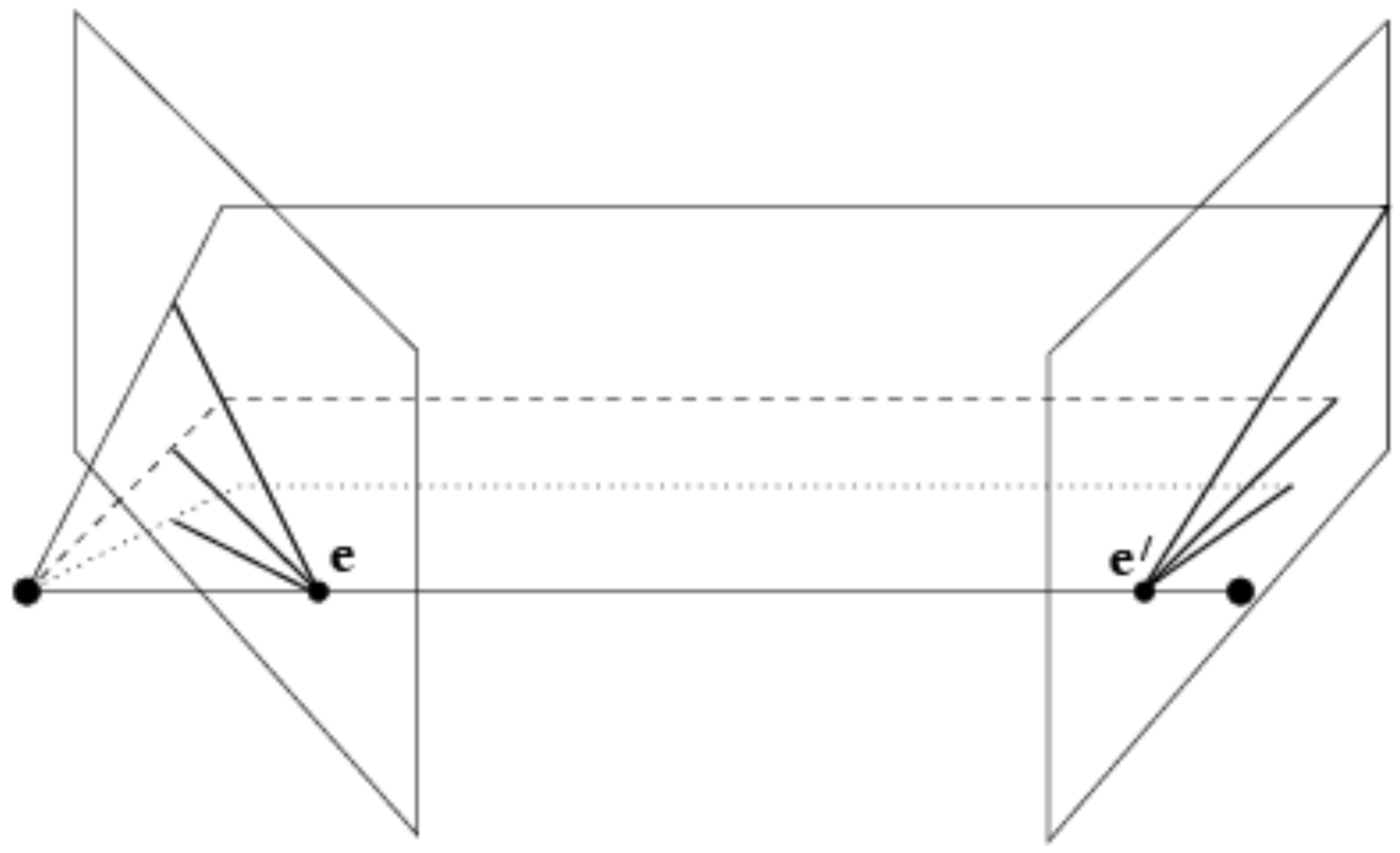
l' is the intersection of this plane with the second image plane

l is the intersection of this plane with the first image plane

The plane is unchanged as we move x along l, or x' along l'

Epipolar Planes and Corresponding Lines



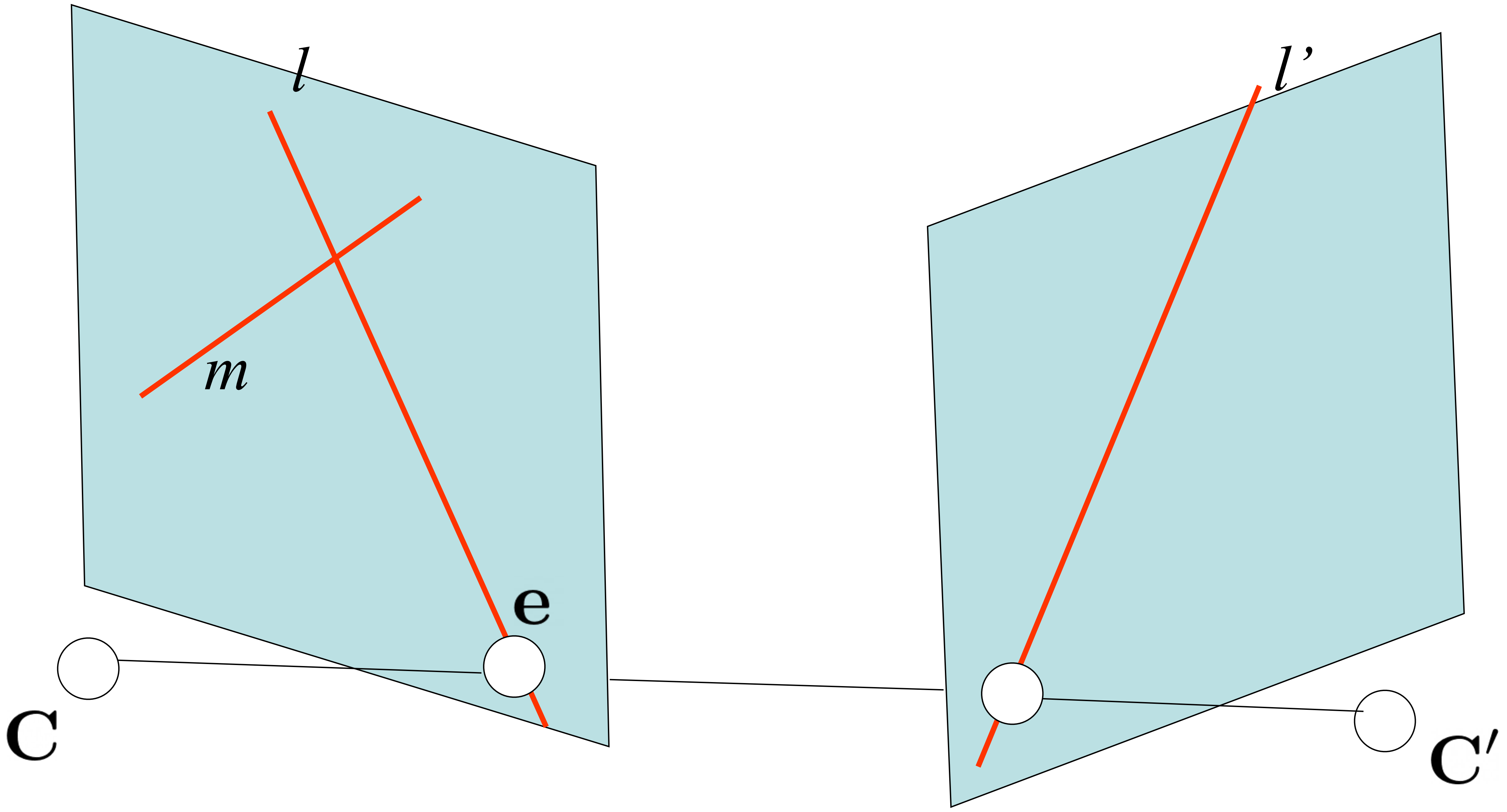


Epipolar planes — set of planes passing **C, C'**

Each epipolar plane intersects the image planes in pairs or corresponding lines **l**, **l'**

Image credits: Martial Hebert, Hartley and Zisserman

Corresponding Epipolar Lines



Given **l**, what is **l**' ?

Consider any line **m** (not passing **e**)

$$\mathbf{l}' = F(\mathbf{m} \times \mathbf{l})$$

why?

A convenient choice: **m**=**e**

$$\mathbf{l}' = F[\mathbf{e}]_\times \mathbf{l}$$

Similarly:

$$\mathbf{l} = F^\mathsf{T}[\mathbf{e}']_\times \mathbf{l}'$$

Fundamental Matrix: cheat sheet

Point Correspondences:

$$\mathbf{x}'^{\top} F \mathbf{x} = 0$$

$$\forall \mathbf{X}, \quad \mathbf{X}^{\top} \underline{\mathbf{P}'^{\top} F \mathbf{P}} \, \mathbf{X} = 0$$

Skew symmetric

Computing F:

$$= [\mathbf{e}']_{\times} \mathbf{P}' \mathbf{P}^{+}$$ given **P**, **P**'

$$= [\mathbf{e}']_{\times} H_{\pi}$$ given **e**' and a homography of an arbitrary plane

Epipoles:

$$\mathbf{e} = \mathbf{P}\mathbf{C}' \qquad \mathbf{e}' = \mathbf{P}'\mathbf{C}$$

$$F\mathbf{e} = \mathbf{0} \qquad \mathbf{e}'^{\top} F = \mathbf{0}^{\top}$$

Epipolar Lines:

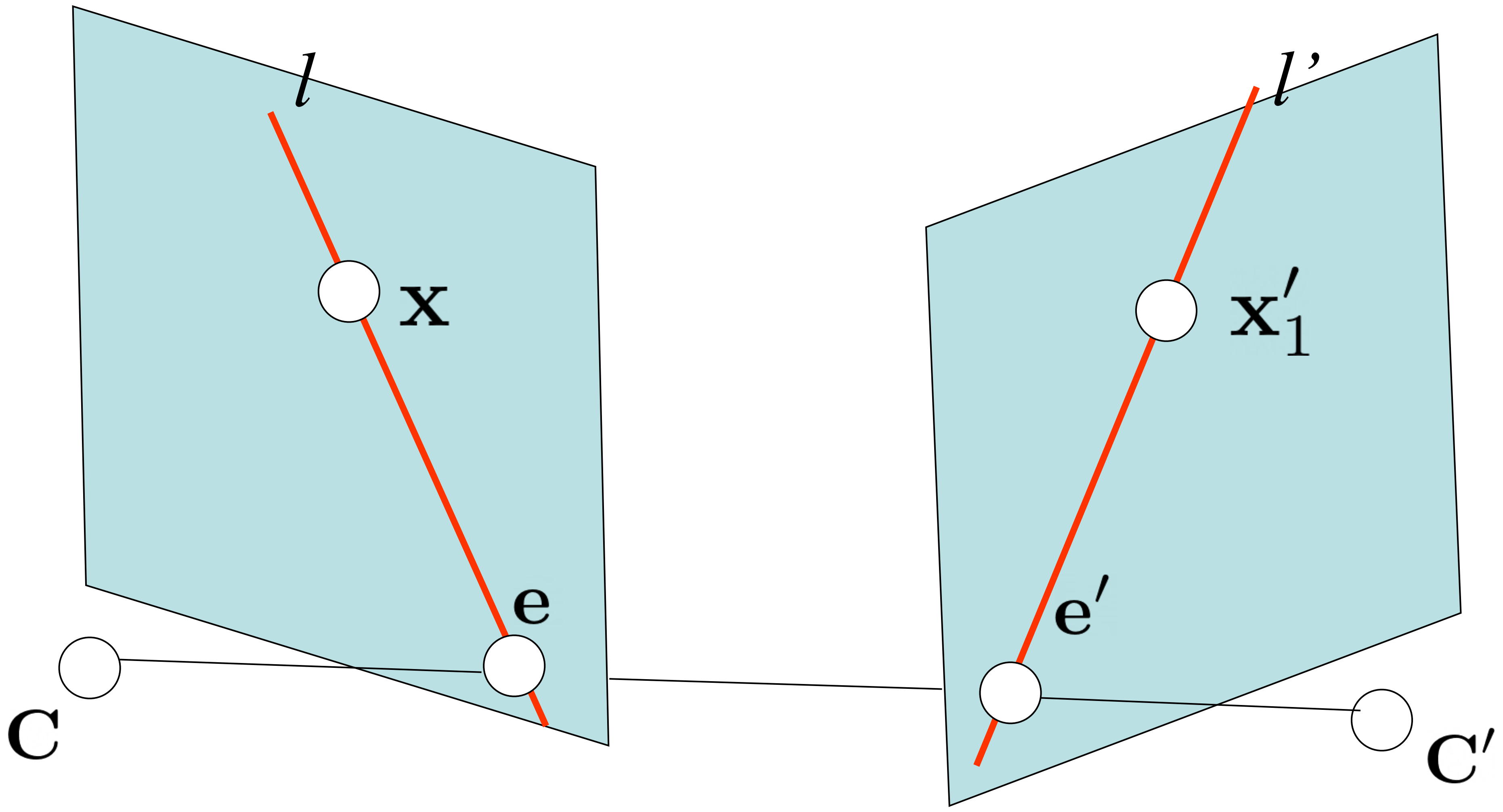
$$\mathbf{l}' = F\mathbf{x} \qquad \mathbf{l} = F^{\top}\mathbf{x}'$$

$$\mathbf{l}' = F[\mathbf{e}]_{\times}\mathbf{l} \qquad \mathbf{l} = F^{\top}[\mathbf{e}']_{\times}\mathbf{l}'$$

Rank and DoF:

$$F = [\mathbf{f}_1 \quad \mathbf{f}_2 \quad \alpha\mathbf{f}_1 + \beta\mathbf{f}_2]$$ (rank 2, 7 DoF)

Fundamental Matrix from Rotation and Translation



$$\mathtt{P} = \mathtt{K}[\mathtt{I} \mid \mathbf{0}] \qquad \mathtt{P}' = \mathtt{K}'[\mathtt{R} \mid \mathbf{t}]$$

$$\mathtt{F} = [\mathtt{P}'\mathbf{C}]_\times \mathtt{P}'\mathtt{P}^+ \qquad \mathtt{P}^+ = \begin{bmatrix} \mathtt{K}^{-1} \\ \mathbf{0}^\mathsf{T} \end{bmatrix} \qquad \mathbf{C} = \begin{pmatrix} \mathbf{0} \\ 1 \end{pmatrix}$$

$$\mathtt{F} = [\mathtt{K}'\mathbf{t}]_\times \mathtt{K}'\mathtt{R}\mathtt{K}^{-1} = \mathtt{K}'^{-\mathsf{T}}[\mathbf{t}]_\times \mathtt{R}\mathtt{K}^{-1}$$

$[\mathbf{t}]_\times \mathtt{M} = \mathtt{M}^{-\mathsf{T}}[\mathtt{M}^{-1}\mathbf{t}]_\times$ *(up to scale)*

Special Case: Rotation

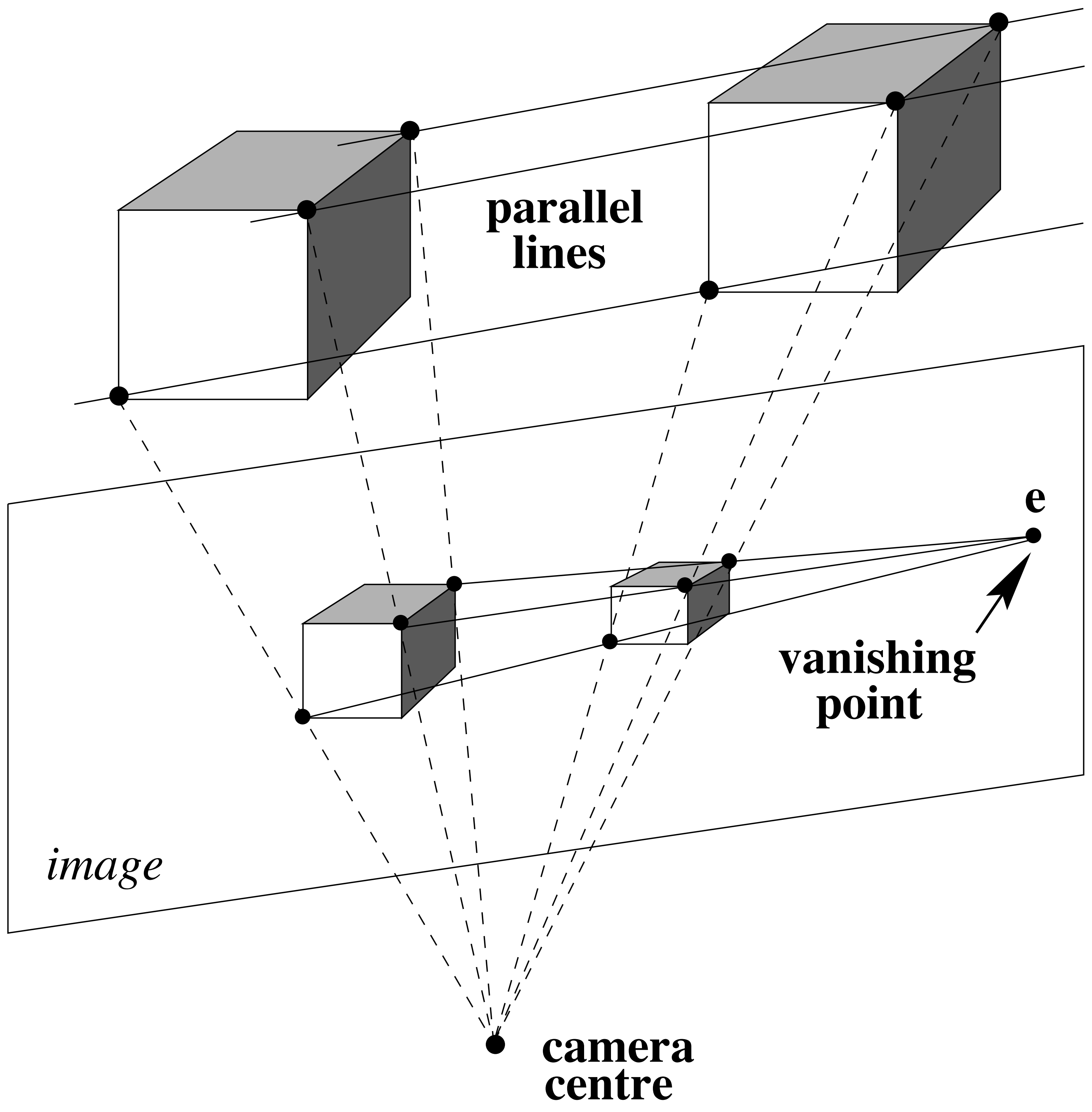
$$\mathbf{x}' = KRK^{-1}\mathbf{x}$$

No valid fundamental matrix, as we get
precise point correspondence!

$$\mathtt{F} = \underbrace{[\mathtt{K}'\mathbf{t}]_{\times}}_{\mathbf{0}\text{ vector}}\mathtt{K}'\mathtt{R}\mathtt{K}^{-1}$$

Important: need to filter out such pairs of images
when computing SfM (as no 3D can be inferred)

Special Case: Translation (R=I, K=K')

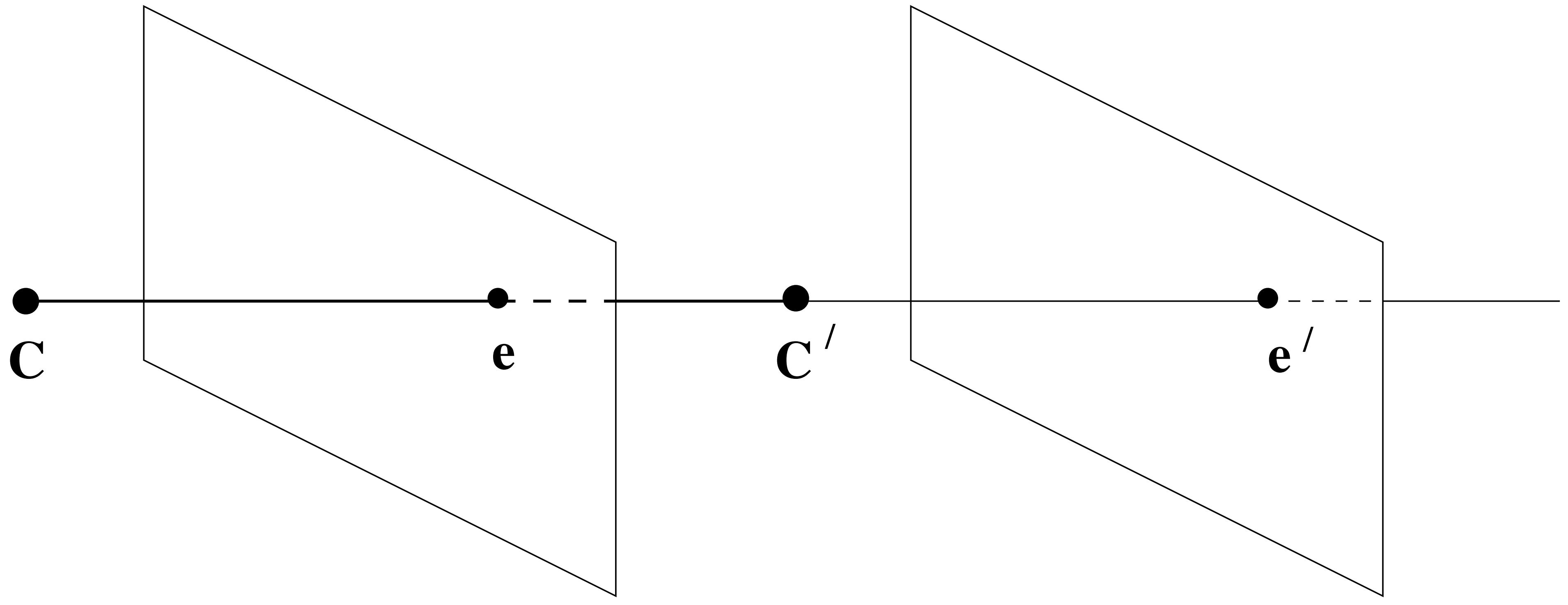


(similar to a static camera, and moving world)

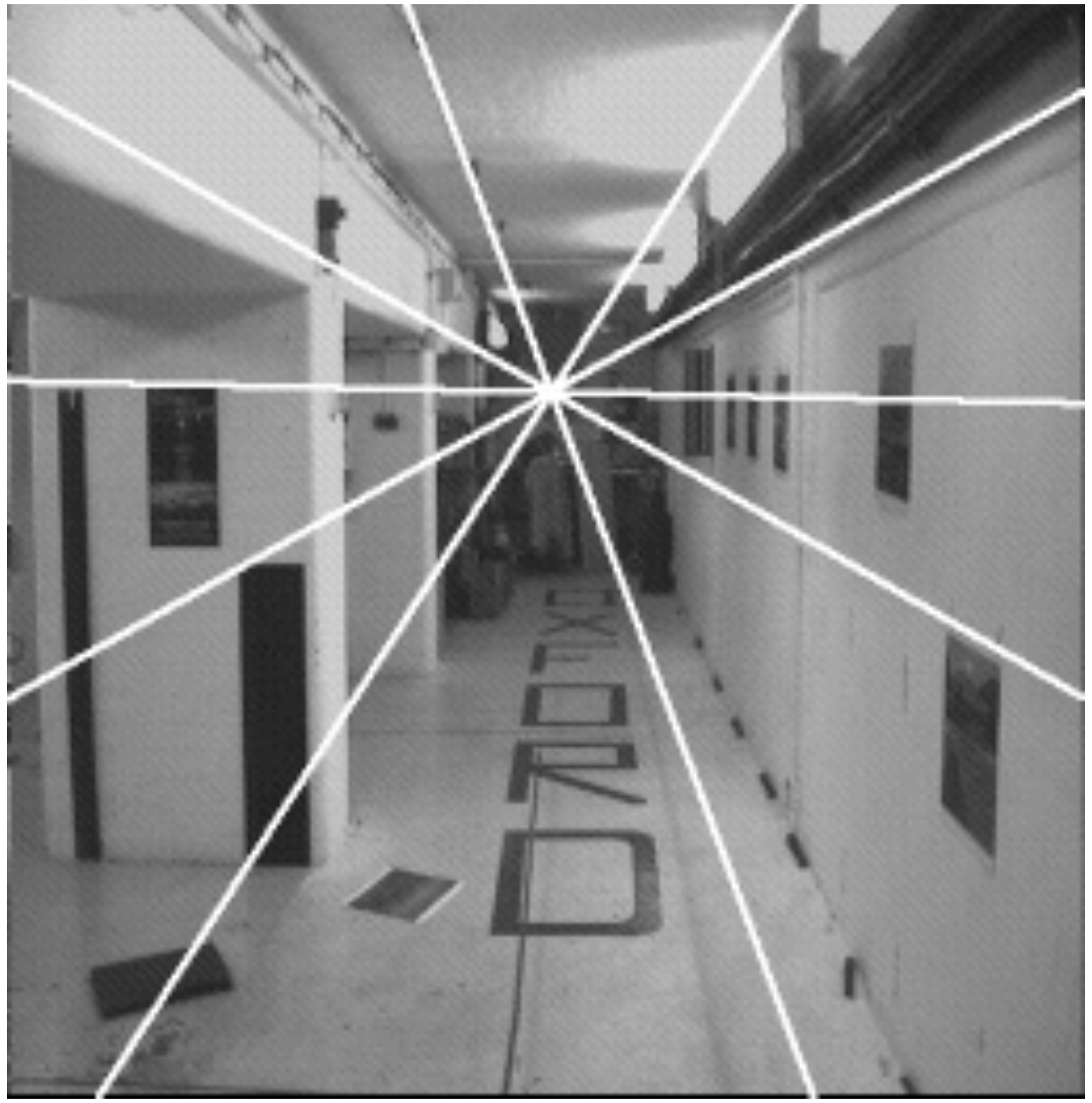
$$\mathtt{F} \;=\; [\mathtt{K}'\mathbf{t}]_\times \mathtt{K}'\mathtt{R}\mathtt{K}^{-1}$$

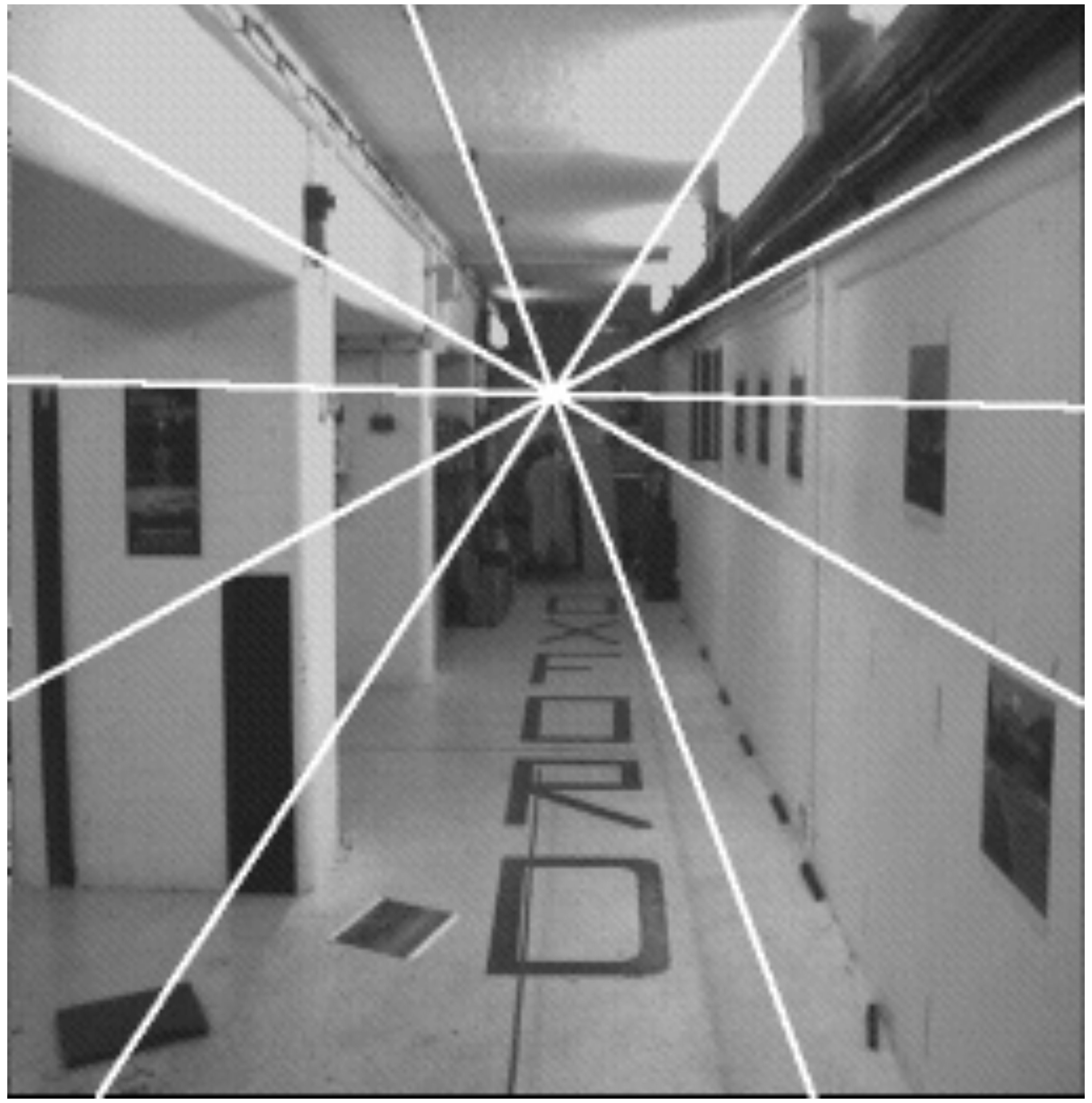
$$F = [\mathbf{e}']_\times$$

$$\mathbf{e}' = K\begin{bmatrix} I & \mathbf{t}\end{bmatrix}\begin{bmatrix}\mathbf{0}\\1\end{bmatrix} = K\mathbf{t} \qquad \mathbf{e} = K\begin{bmatrix} I & \mathbf{0}\end{bmatrix}\begin{bmatrix}\mathbf{t}\\1\end{bmatrix} = K\mathbf{t}$$



Special Case: Translation (R=I, K=K’)

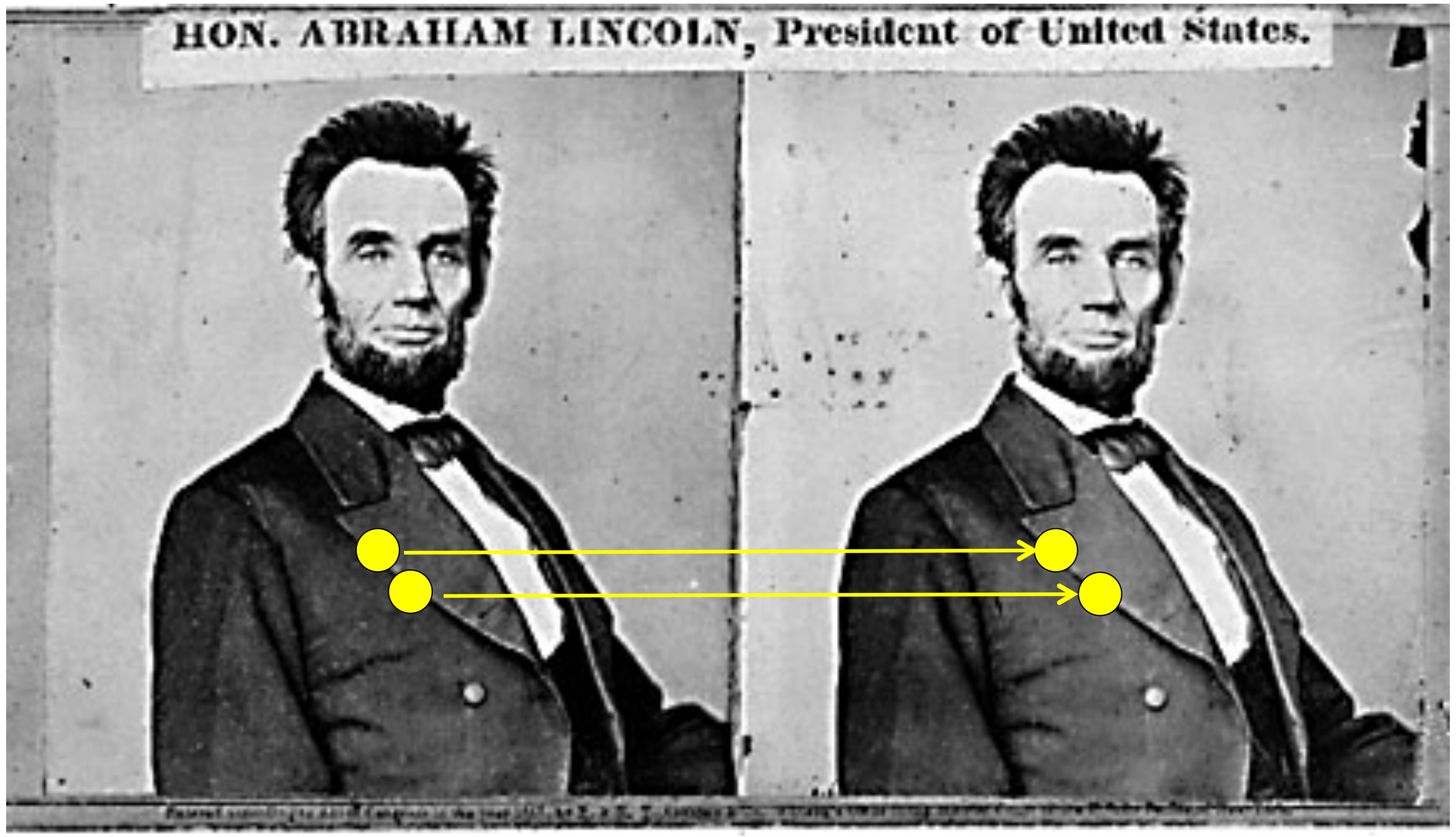




$$\mathbf{e}' = K \begin{bmatrix} I & \mathbf{t} \end{bmatrix} \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix} = K\mathbf{t}$$

$$\mathbf{e} = K \begin{bmatrix} I & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{t} \\ 1 \end{bmatrix} = K\mathbf{t}$$

Special Case: Binocular Stereo



Q: without any math, tell the coordinates of the epipole?

Essential Matrix: Relating Normalized Coordinates

$$\mathbf{x}'^{\top} F \mathbf{x} = 0$$

$$\hat{\mathbf{x}}'^{\top} E \hat{\mathbf{x}} = 0$$

$$\hat{\mathbf{x}} = \mathbf{K}^{-1}\mathbf{x}, \ \hat{\mathbf{x}}' = \mathbf{K}'^{-1}\mathbf{x}'$$

Just as F relates image coordinates, E relates ‘normalized’ coordinates

$$\mathtt{E} = \mathtt{K}'^{\top}\mathtt{F}\mathtt{K}$$

Given two cameras related by **R, t** :

$$\mathtt{F} = \mathtt{K}'^{-\top}[\mathbf{t}]_{\times}\mathtt{R}\mathtt{K}^{-1}$$

$$\mathtt{E} = [\mathbf{t}]_{\times}\mathtt{R}$$

Q: How many degrees of freedom?

Questions?